Antoni Kowalski

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SUMMARY

Senior Software Engineer with 8+ years building production systems across cybersecurity, fintech, healthcare, SaaS, and large-scale consumer platforms.
Expertise in **cybersecurity AI, large language models, Python, PyTorch, and Kubernetes** for secure reasoning, summarization, interactivity, and threat investigation.
Proven track record delivering **research-to-production machine learning systems** with responsible AI, adversarial evaluation, model observability, and data protection controls.
Proficient in deep learning training and deployment at scale, NVIDIA GPU optimization, distributed systems, cloud technologies, and production-grade model evaluation frameworks.
Collaborative technical leader experienced mentoring engineers, guiding cross-functional initiatives, influencing senior stakeholders, and making high-judgement decisions in ambiguous, fast-paced environments.
Strong focus on **LLM architecture, cybersecurity applications, engineering excellence, and scalable AI systems** that improve detection, analyst workflows, and customer protection.

SKILLS

Programming & Scientific Computing: Python, C++, SQL, Bash, NumPy, Pandas, SciPy, Jupyter, Pydantic, FastAPI

LLM & Generative AI: Large Language Models, Transformers, Retrieval-Augmented Generation, Prompt Engineering, Supervised Fine-Tuning, Parameter-Efficient Fine-Tuning, Reinforcement Learning From Human Feedback, Agentic AI, Embeddings, Vector Search

Deep Learning & Machine Learning: PyTorch, TensorFlow, Scikit-Learn, Deep Learning, Natural Language Processing, Representation Learning, Distributed Training, Model Calibration, Feature Engineering, Hyperparameter Optimization

Data & Distributed Systems: Apache Kafka, Apache Spark, PostgreSQL, Redis, Elasticsearch, Milvus, Pgvector, Ray, gRPC, REST APIs

EXPERIENCE

Senior Software Engineer — Bounteous Mar 2024 – Present

Warsaw, Poland (Remote)

- Architected a **production-grade retrieval-augmented generation platform** using Python, PyTorch, Transformers, vector search, and Kubernetes, enabling secure reasoning and summarization across sensitive enterprise cybersecurity knowledge bases.
- Led research strategy for LLM evaluation, prompt optimization, and fine-tuning, translating ambiguous product requirements into reproducible experiments, deployment criteria, technical standards, and measurable model-quality benchmarks.
- Engineered **distributed inference services for transformer models** with NVIDIA GPUs, mixed precision, dynamic batching, and autoscaling, supporting 120 million security signals daily with predictable latency.
- Designed agentic workflows for threat investigation using tool calling, semantic retrieval, and structured outputs, improving analyst interactivity while preserving authorization boundaries, auditability, and data governance.
- Implemented **adversarial evaluation and AI red-teaming frameworks** covering prompt injection, data leakage, hallucination, and unsafe tool execution for enterprise cybersecurity products and responsible AI adoption.
- Optimized PyTorch inference through quantization, compilation, caching, and GPU memory profiling, reducing end-to-end response time from 11 seconds to 4 seconds across six production models.

- Mentored engineers and applied scientists on deep learning experimentation, model review, and research-to-production practices while aligning cross-functional technical strategy with senior product and security stakeholders.
- Established **LLM observability and continuous evaluation pipelines** using MLflow, OpenTelemetry, and statistical test suites, enabling safe releases, drift detection, incident analysis, and production rollback decisions.

Senior Software Engineer — Wise Oct 2022 – Feb 2024
Warsaw, Poland (Remote)

- Developed **machine learning systems for payment risk detection** using Python, PyTorch, Kafka, and PostgreSQL, combining behavioral signals and transaction context for secure, real-time financial decisioning.
- Designed deep learning training pipelines for fraud classification, feature representation, and model calibration, integrating reproducible experiments, offline evaluation, and controlled production deployment across global fintech services.
- Built **streaming feature pipelines and online inference APIs** processing more than 2 million daily payment events while maintaining low-latency decisions, traceability, and resilient fallback behavior.
- Implemented model explainability, bias analysis, and monitoring for regulated financial workflows, supporting audit evidence, GDPR-aligned data handling, and high-judgement decisions by experienced risk operations teams.
- Optimized GPU training with mixed precision, distributed data parallelism, and efficient sampling, shortening a core model training cycle from 19 hours to 7 hours while preserving accuracy.
- Partnered with software engineers and risk specialists to translate emerging fraud patterns into **production-grade model architectures**, evaluation datasets, deployment controls, and documented incident response playbooks.
- Established technical standards for feature quality, experiment tracking, and model validation, improving consistency across applied machine learning projects supporting payment processing, fraud detection, and customer protection.
- Mentored engineers through architecture reviews and model debugging, communicating tradeoffs to senior stakeholders and driving cross-functional decisions under ambiguity within a fast-paced global fintech environment.

EDUCATION

University of Oulu — Bachelor’s degree Aug 2014 – Jun 2017
Oulu, Finland (Onsite)